

Digital Image Processing

Texture Analysis

Texture Based Descriptors

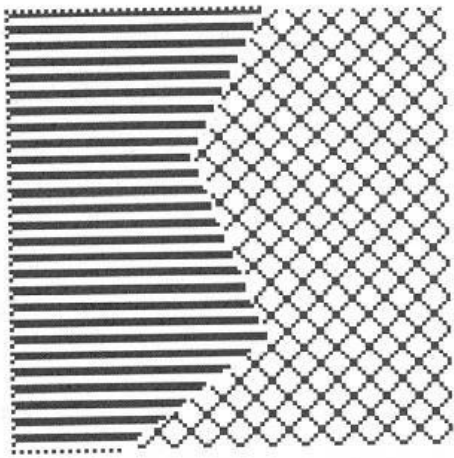
Texture

- Organized patterns of quite regular subelements called textons.
- Texture is a property of sufficiently large regions

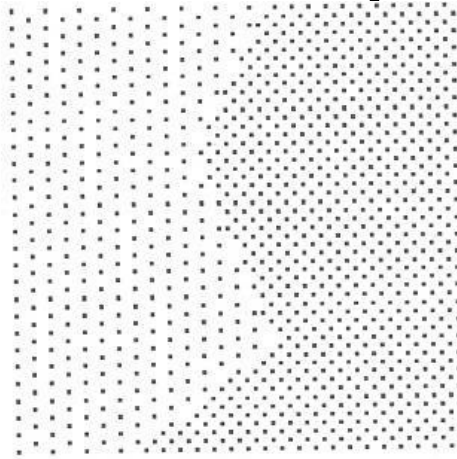
Applications:

- Texture based segmentation
- Texture synthesis
- Texture analysis and texture based matching
- Shape (surface orientation) from texture

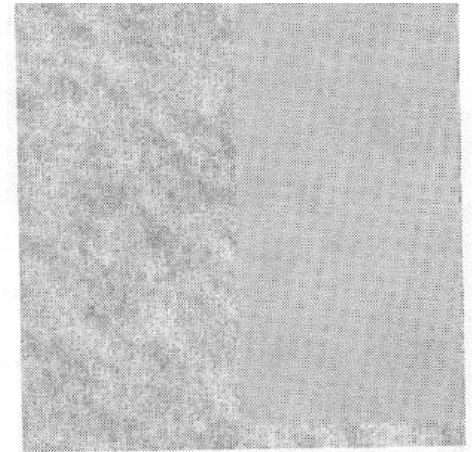
Texture Examples



Test image T1
(a)



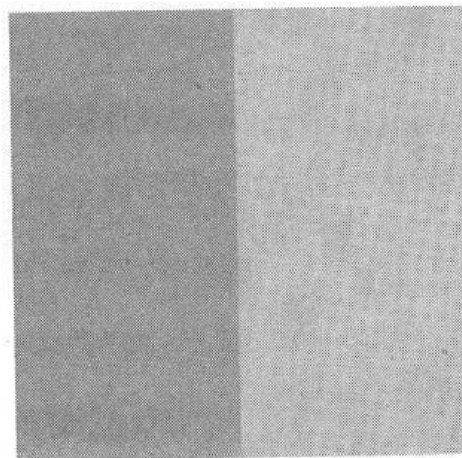
Test image T2
(b)



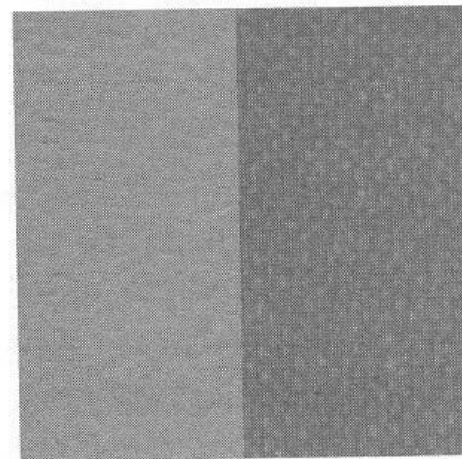
Test image T3
(c)

(a,b): Artificial
textures

(c,d,e): Naturally
occurring
textures



Test image T4
(d)



Test image T5
(e)

Representing

Statistical textures

yields characterization of textures as smooth, coarse grainy, etc.

Spectral

are based on Fourier spectrum and are primarily used to detect the global periodicity in an image by identifying high energy narrow peaks in the spectrum.

Statistical approaches

- Based on the histogram measures of image
- Based on the Grey Level Co-occurrence Matrix (GLCM) and related measurement

Histogram based texture description

Using statistical moments of grey level histogram of the image or region

Let $p(z_i)$ is the histogram of the grey levels z_i of an image

The nth moment about the mean is given by:

$$\mu_n(z) = \frac{1}{L} \sum_{i=0}^{L-1} (z_i - m)^n p(z_i)$$

Where mean is

$$m = \sum_{i=0}^{L-1} z_i p(z_i)$$

The variance is the second moment and is given by

$$\sigma^2(z) = \mu_2(z) - \mu^2(z) = \frac{1}{L} \sum_{i=0}^{L-1} (z_i - m)^2 p(z_i) - \left(\frac{1}{L} \sum_{i=0}^{L-1} z_i p(z_i) \right)^2$$

Histogram based texture description

- For texture description the following parameters are useful

- Variance and related measures: descriptor of relative smoothness, use normalized variance

$$R = 1 - \frac{1}{1 + \sigma^2(z)}$$

- Skewness of histogram

$$\mu_3(z) = \sum_{i=0}^L (z_i - \bar{m})^3 p(z_i)$$

- Relative flatness of histogram

$$\mu_4(z) = \sum_{i=0}^L (z_i - \bar{m})^4 p(z_i)$$

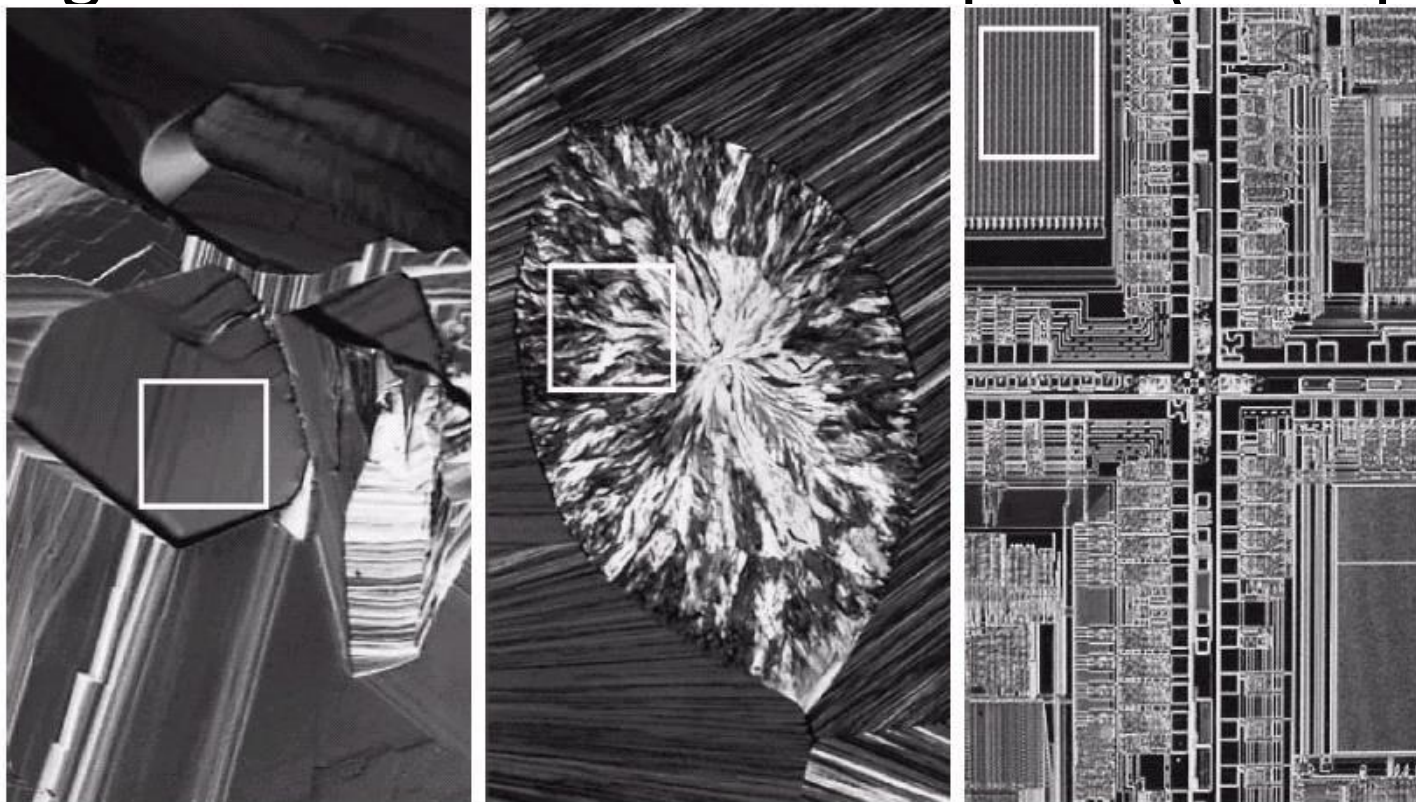
- Uniformity

$$U = \sum_{i=0}^{L-1} p^2(z_i)$$

- Average Entropy

$$e = - \sum_{i=0}^L p(z_i) \log_2 p(z_i)$$

Histogram based texture description (example)



Texture	Mean	Standard deviation	R (normalized)	Third moment	Uniformity	Entropy
Smooth	82.64	11.79	0.002	-0.105	0.026	5.434
Coarse	143.56	74.63	0.079	-0.151	0.005	7.783
Regular	99.72	33.73	0.017	0.750	0.013	6.674

GLCMs

- For texture description the following parameters of GLCM are measured and analyzed

Maximum probability

$$\max_{i,j} (c_{ij})$$

Contrast

$$\sum_i \sum_j (i - j)^2 c_{ij}$$

Uniformity

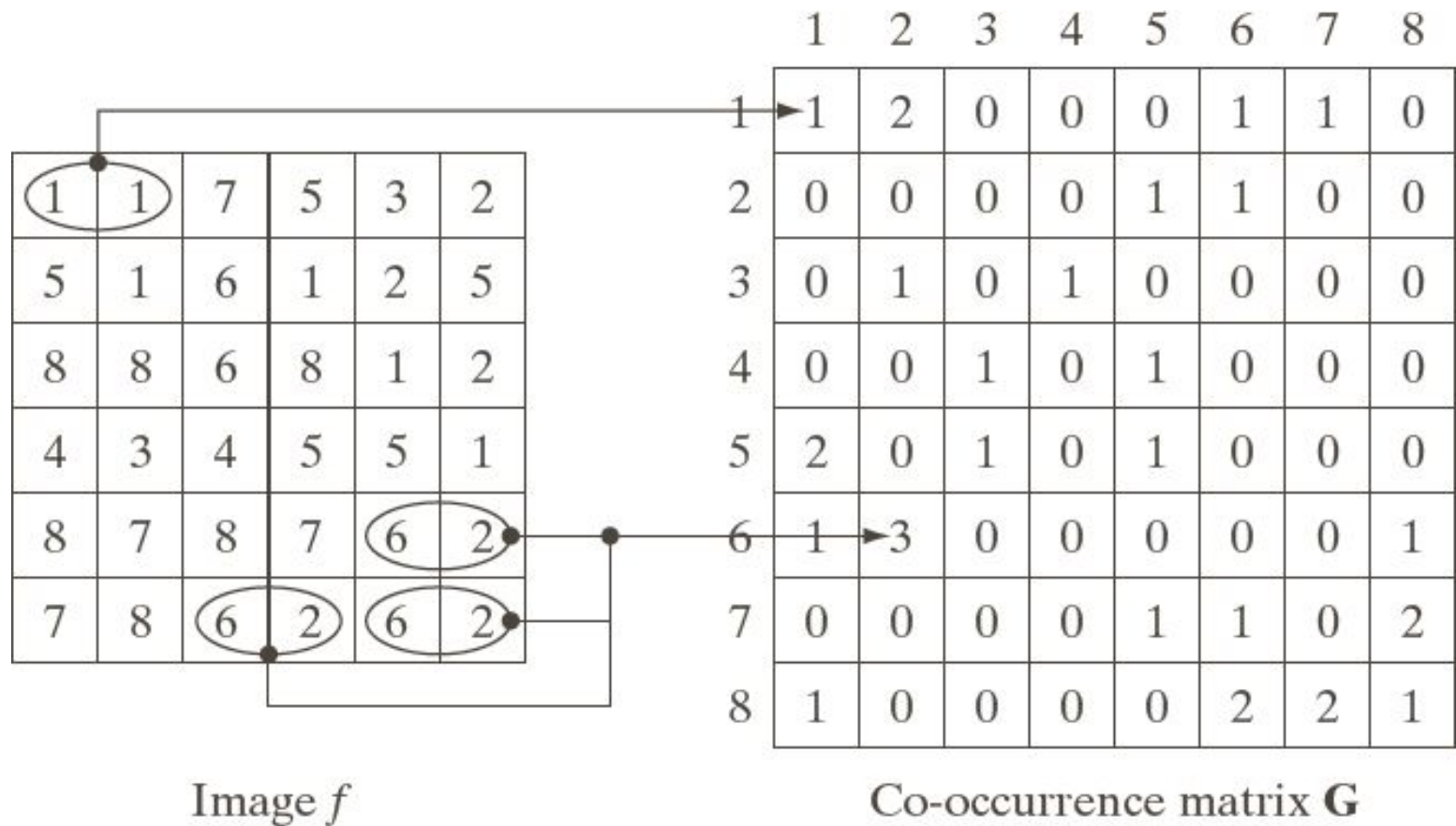
$$\sum_i \sum_j c_{ij}^2$$

Entropy

$$-\sum_{i,j} c_{ij} \log c_{ij}$$

FIGURE 11.29

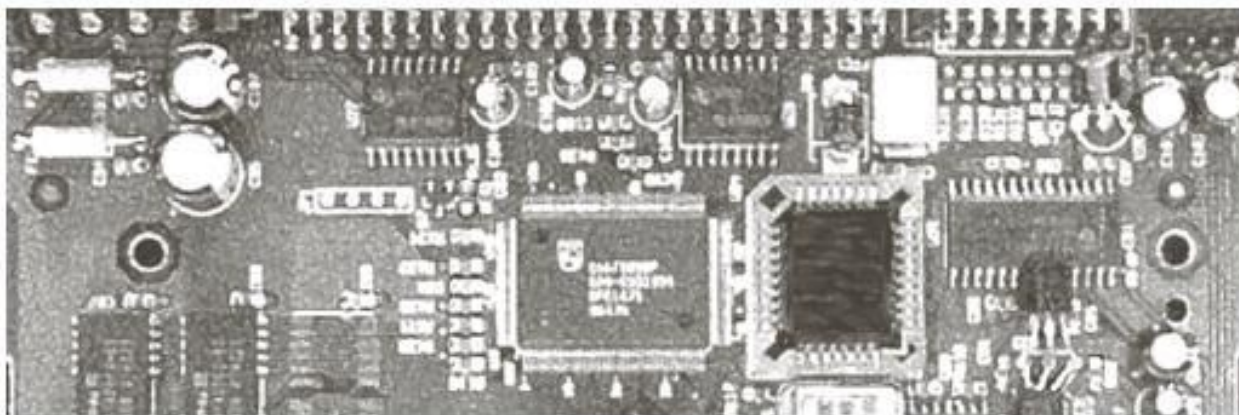
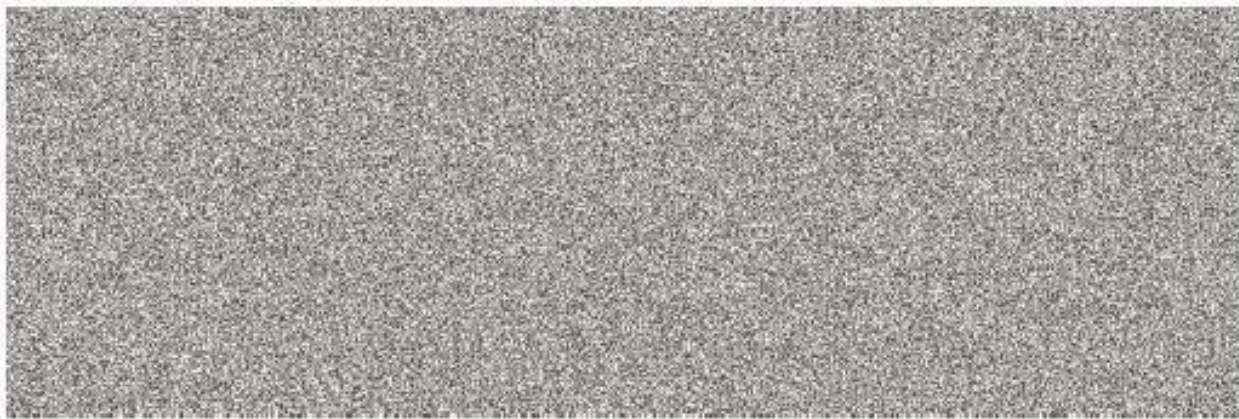
How to generate
a co-occurrence
matrix.



Descriptor	Explanation	Formula
Maximum probability	Measures the strongest response of G . The range of values is [0, 1].	$\max_{i,j}(p_{ij})$
Correlation	A measure of how correlated a pixel is to its neighbor over the entire image. Range of values is 1 to -1, corresponding to perfect positive and perfect negative correlations. This measure is not defined if either standard deviation is zero.	$\sum_{i=1}^K \sum_{j=1}^K \frac{(i - m_r)(j - m_c)p_{ij}}{\sigma_r \sigma_c}$ $\sigma_r \neq 0; \sigma_c \neq 0$
Contrast	A measure of intensity contrast between a pixel and its neighbor over the entire image. The range of values is 0 (when G is constant) to $(K - 1)^2$.	$\sum_{i=1}^K \sum_{j=1}^K (i - j)^2 p_{ij}$
Uniformity (also called Energy)	A measure of uniformity in the range [0, 1]. Uniformity is 1 for a constant image.	$\sum_{i=1}^K \sum_{j=1}^K p_{ij}^2$
Homogeneity	Measures the spatial closeness of the distribution of elements in G to the diagonal. The range of values is [0, 1], with the maximum being achieved when G is a diagonal matrix.	$\sum_{i=1}^K \sum_{j=1}^K \frac{p_{ij}}{1 + i - j }$
Entropy	Measures the randomness of the elements of G . The entropy is 0 when all p_{ij} 's are 0 and is maximum when all p_{ij} 's are equal. The maximum value is $2 \log_2 K$. (See Eq. (11.3-9) regarding entropy).	$-\sum_{i=1}^K \sum_{j=1}^K p_{ij} \log_2 p_{ij}$

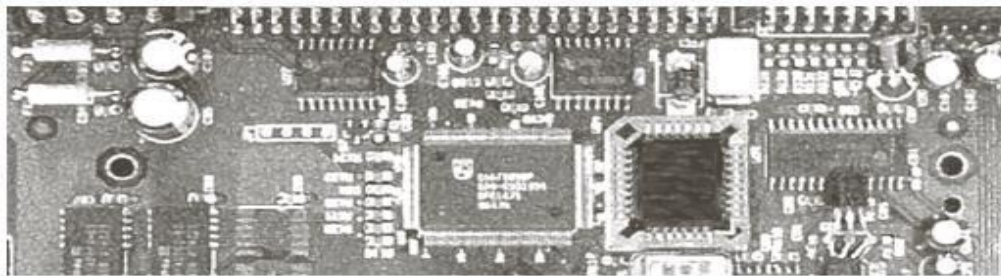
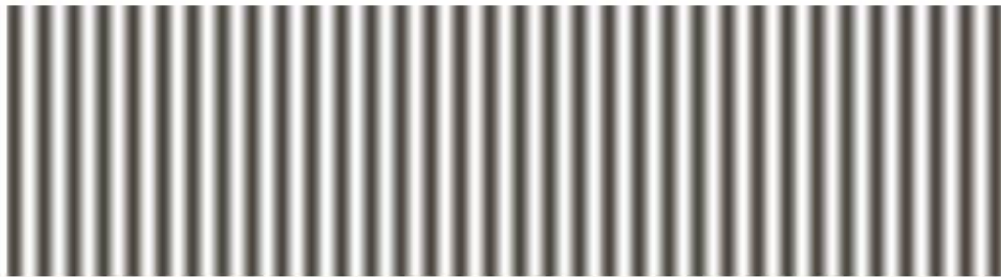
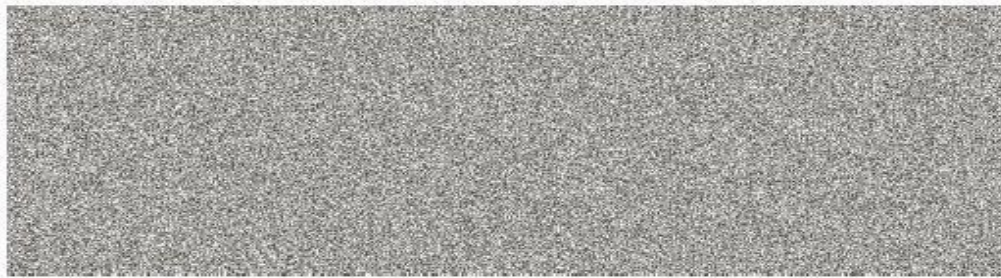
TABLE 11.3

Descriptors used for characterizing co-occurrence matrices of size $K \times K$. The term p_{ij} is the ij th term of **G** divided by the sum of the elements of **G**.



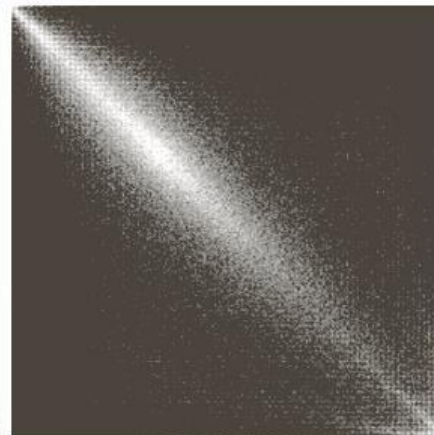
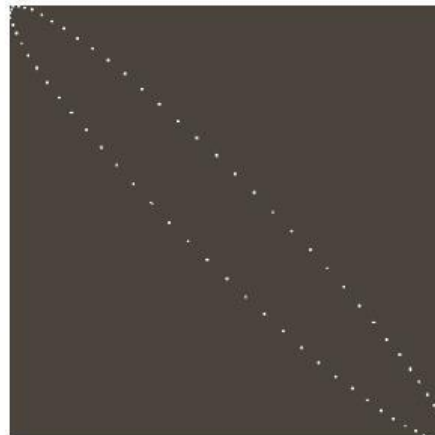
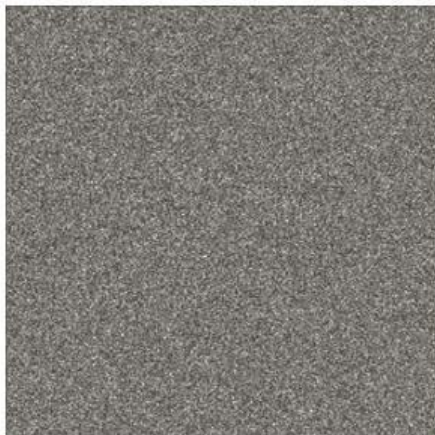
a
b
c

FIGURE 11.30
Images whose
pixels have
(a) random,
(b) periodic, and
(c) mixed texture
patterns. Each
image is of size
 263×800 pixels.



a b c

FIGURE 11.31
 256×256 co-occurrence matrices, G_1 , G_2 , and G_3 , corresponding from left to right to the images in Fig. 11.30.



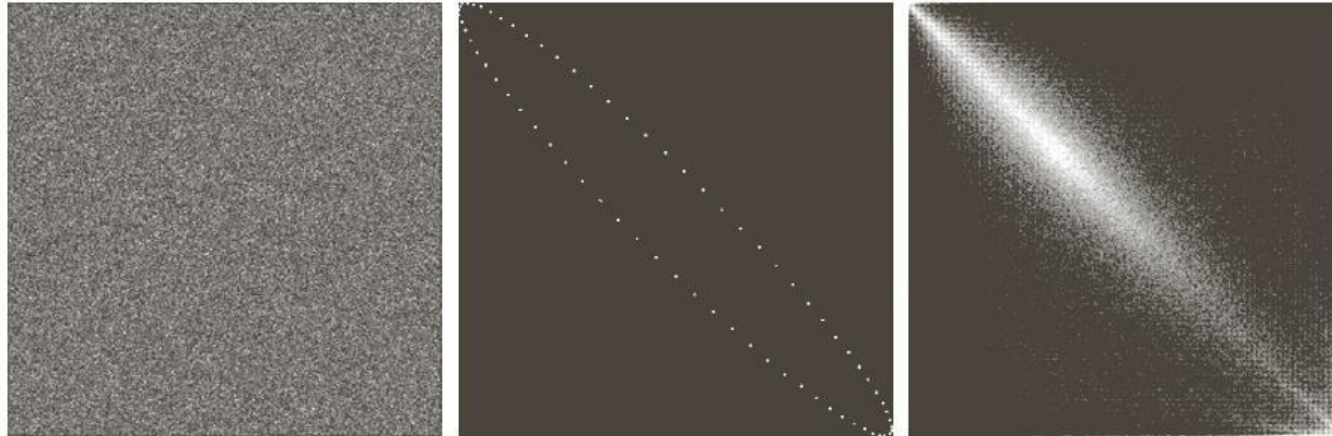


TABLE 11.4
Descriptors
evaluated using
the co-occurrence
matrices displayed
in Fig. 11.31.

Normalized Co-occurrence Matrix	Descriptor					
	Max Probability	Correlation	Contrast	Uniformity	Homogeneity	Entropy
$\mathbf{G}_1/n_1$	0.00006	-0.0005	10838	0.00002	0.0366	15.75
$\mathbf{G}_2/n_2$	0.01500	0.9650	570	0.01230	0.0824	6.43
$\mathbf{G}_3/n_3$	0.06860	0.8798	1356	0.00480	0.2048	13.58

Acknowledgements

- ◆ Digital Image Processing”, Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2002
- ◆ Computer Vision for Computer Graphics, Mark Borg